LESSON 2.12

Multiply Decimals

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can multiply decimals and place the decimal point correctly in the product.

WHY THIS MATTERS The station needs 4.5 meters of heat shielding that costs $12.60 per meter. Commander Diaz needs to calculate the total cost before submitting the supply order. What is the total cost of the shielding?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Product *Spanish: Producto*	The answer when you multiply.	4 × 7 = 28, so the product is 28
Decimal point *Spanish: Punto decimal*	The dot that splits the whole number from the part after it.	In 3.25, the decimal point separates 3 (ones) from 25 (tenths and hundredths)
Estimate *Spanish: Estimar*	A close answer you get by rounding first.	3.7 × 4.2: round to 4 × 4 = 16, so the exact answer should be near 16
Place value *Spanish: Valor posicional*	What a digit is worth based on where it sits in a number.	In 0.36: the 3 is in the tenths place (0.3) and the 6 is in the hundredths place (0.06)
Decimal places *Spanish: Cifras decimales*	How many digits come after the decimal point.	3.4 (1 place) × 2.6 (1 place) = 8.84 (2 places total)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want 4.5 × 12.60, the cost of 4.5 meters at $12.60 each.

1. First I multiply without points, like whole numbers: 45 × 1260 = 56,700.
2. Now I count decimal places: 4.5 has 1, and 12.60 has 2, so 1 + 2 = 3 decimal places.
3. I put the point 3 places from the right in 56,700: that gives 56.700, or $56.70.
4. I check with an estimate: 5 × $13 = $65, and $56.70 is close, so my answer is reasonable.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's multiply 0.4 × 0.7. First multiply like whole numbers: what is 4 × 7? (28.)

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I estimated _____ x _____ by rounding to _____ x _____ = _____, which told me the product is _____ than 20.

Word bank Product | Decimal point | Estimate | Place value | Decimal places | estimate | round | decimal places | product | multiply | reasonable

WATCH OUT!

A common mistake

A common mistake in Multiply Decimals is counting decimal places from only one factor instead of adding the decimal places from BOTH factors. For example, in 0.4 × 0.06, a student multiplies 4 × 6 = 24, then places the decimal point using just one factor's place count (1 place), writing 0.24 instead of the correct answer: 0.4 has 1 decimal place and 0.06 has 2 decimal places, for a total of 3, so 24 becomes 0.024. Before you submit, count the decimal places in EACH factor separately and add them together

LESSON 2.12

Multiply Decimals

Version A · Supported practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Multiply as whole numbers, then place the decimal point.

Problem	Whole Number Product	Total Decimal Places	Final Answer
2.5 × 4			
1.2 × 0.3			
6.1 × 0.5			

HINT The columns show the whole strategy in order: whole-number product first, then count decimal places, then place the point.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 3.4 × 2.6?

(A) 0.884
(B) 8.48
(C) 8.84
(D) 88.4

HINT Decimal multiplication is two jobs: multiply the digits, then place the point. Count the decimal places in BOTH factors first.

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

3 **Circle the letter of the best answer. Show how you know.**

What is 0.7 × 1.5?

(A) 0.105
(B) 1.05
(C) 1.50
(D) 10.5

HINT Both factors are close to 1, so expect an answer near 1 — a quick estimate protects you from decimal-point slips.

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

4 **Circle the letter of the best answer. Show how you know.**

What is 5.3 × 4?

(A) 2.12
(B) 20.12
(C) 21.2
(D) 212

HINT Only ONE factor has a decimal here: 5.3 has one place and 4 has none, so the product needs exactly one decimal place.

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

PART 3 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Which estimate is closest to 4.8 × 3.2?

(A) 4 × 3 = 12
(B) 5 × 3 = 15
(C) 5 × 4 = 20
(D) 48 × 32 = 1,536

HINT 'Estimate' means round each factor to its nearest whole number BEFORE multiplying — no exact work needed.

SHOW YOUR WORK

PART 4 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — Counting Decimal Places

1. **Problem** 0.3 × 0.6
2. **Multiply as whole numbers** 3 × 6 = 18
3. **Count decimal places** 0.3 has 1 decimal place and 0.6 has 1 decimal place → 1 total
4. **Place the decimal point** Move the point 1 place in 18 to get 1.8

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Add the decimal places in BOTH factors: how many does 0.3 have, and how many does 0.6 have?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did estimating by rounding help you sort the products? How do you decide where to place the decimal in the exact product?

SENTENCE STARTER I estimated ___ x ___ by rounding to ___ x ___ = ___, which told me the product is ___ than 20.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.12

Multiply Decimals

Version B · Core practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each expression to its product.

_____ 2.5 × 4	(A) 10
_____ 1.2 × 0.3	(B) 0.36
_____ 3.6 × 0.5	(C) 1.8
_____ 4.8 × 2.5	(D) 12
_____ 0.6 × 0.8	(E) 0.48
_____ 7.5 × 1.2	(F) 9

2 **Complete the table. Show how you found each value.**

Complete each decimal multiplication.

Problem	Product
6.4 × 3.5	
0.25 × 0.4	
8.3 × 2.1	
12.5 × 0.8	

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Sort: when you multiply two decimals, is the product GREATER than both factors, or LESS than at least one factor?

Groups: **Product Greater Than Both Factors** | **Product Less Than at Least One Factor**

_____ 3.5 × 4

_____ 6 × 2.5

_____ 10 × 1.5

_____ 0.5 × 8

_____ 0.3 × 0.4

_____ 0.7 × 6

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A car travels 55.5 miles per hour for 2.4 hours. How far does it go?

(A) 13.32 miles

(B) 57.9 miles

(C) 133.2 miles

(D) 1,332 miles

PER 1

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — Placing the Decimal Point

1. **Problem** 1.5 × 2.4
2. **Multiply as whole numbers** 15 × 24 = 360
3. **Count decimal places** 1.5 has 1 decimal place and 2.4 has 1 decimal place → 2 total
4. **Place the decimal point** Move the point 2 places in 360 to get 0.36

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did estimating by rounding help you sort the products? How do you decide where to place the decimal in the exact product?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.12

Multiply Decimals

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can multiply decimals and place the decimal point correctly in the product.

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the missing factor.

Equation	Missing Factor
___ × 0.5 = 3.5	
2.4 × ___ = 7.2	
___ × 1.5 = 4.5	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Kai's Mistake

1. **Problem** 2.4 × 3.15
2. **Multiply as whole numbers** 24 × 315 = 7,560
3. **Count decimal places** 2.4 has 1 decimal place, 3.15 has 2 decimal places → 2 total
4. **Place decimal** 75.60

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A driver buys 9.5 gallons of gas at $3.40 per gallon. Show how to find the total cost. Use estimation to check if your answer is reasonable.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did estimating by rounding help you sort the products? How do you decide where to place the decimal in the exact product?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help